48. The distribution of the number of defective blades produced in a single shift in a factory over 100 shifts is given below.

Number of defective blades	0	1	2	3	4
Number of shifts	12	14	23	18	33

Test whether the number of defective blades follows a Poisson distribution with mean = 0.44. Use $\alpha = 0.05$.

49. The quality grade of electric components produced in two factories is given in the table given below.

Factory	Quality of grade				Total
	Poor	Medium	Good	Excellent	
A	136	165	151	148	600
B	31	58	55	36	180
Total	167	223	206	184	780

Test whether there is any association between factories and quality of grades.

50. The eyesight was tested among 2000 randomly selected patients from a city and the following details are obtained.

Gender	Eye-sight		Total
	Poor	Good	
Male	620	380	1000
Female	550	450	1000
Total	1170	830	2000

Can we conclude that there is an association between gender and quality of eye-sight at 5% level of significance?

51. The weights (in kg) of 10 students from a school are 38,40,45,53,47,43,55,48,52,49. Can we say that variance of the distribution of weights of all students from the above school is equal to 20 kg?

52. In a sample of 200 households in a colony, two brands of hair oils A and B are applied by 90 females. Further, 60 females and 70 males are using brand A. To test whether there is any association between the gender and brand of hair oil used, by constructing a contigency table.

12th_Statistics_EM_Unit_2.indd 74 1/28/2020 7:10:17 PM